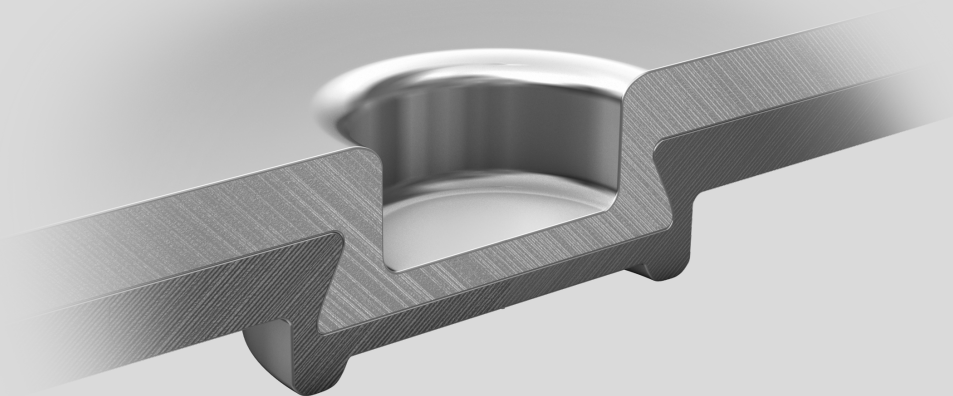


# TOX® Clinching Technology

## Clinching Tools and Accessories

Data sheet 80.07  
2021/07



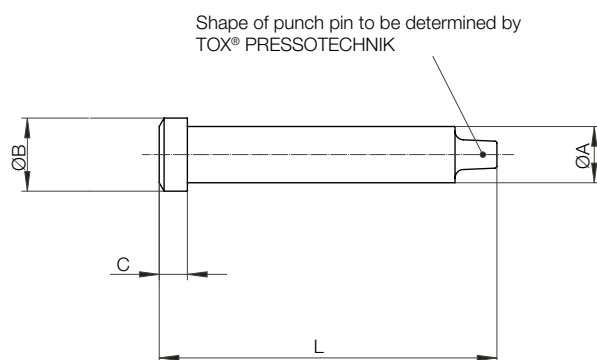
# Tools

## Flanged Tools

The dimensions and shapes of TOX® Clinching Tools are standardized. Besides the standard tool lengths and point diameters listed here, many special solutions are available on request.

Thus, the punches and dies can be adapted to the geometry of the used piece parts. For detailed information please contact our local sales representations.

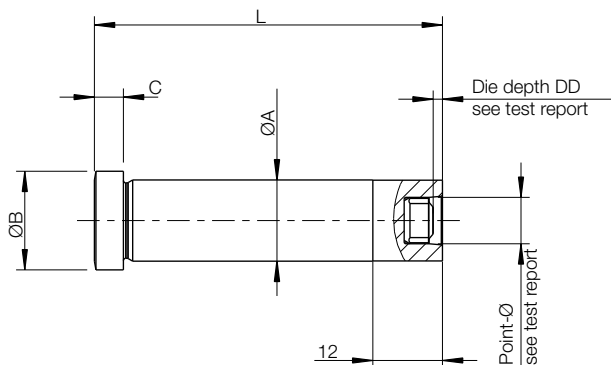
### TOX® Flanged Punch



| Type       | L   | Ø A <sup>-0.005</sup> <sub>-0.020</sub> | Ø B | C | Point Ø |
|------------|-----|-----------------------------------------|-----|---|---------|
| TOX 03.140 | 40  | 3                                       | 5   | 3 | 2       |
| TOX 10.25  | 60  | 10                                      | 13  | 5 | 3 - 12  |
| TOX 10.180 | 80  | 10                                      | 13  | 5 | 3 - 12  |
| TOX 10.200 | 100 | 10                                      | 13  | 5 | 3 - 12  |
| TOX 10.220 | 120 | 10                                      | 13  | 5 | 3 - 12  |
| TOX 14.25  | 60  | 14                                      | 17  | 5 | 6 - 12  |

Dimensions in mm

### TOX® Round Joint flanged die

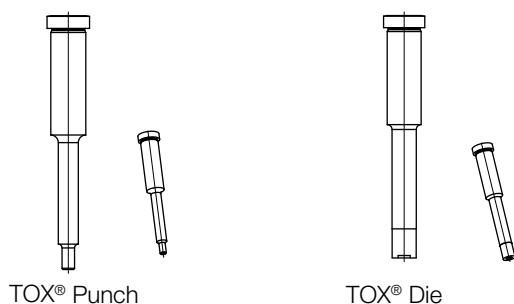


| Type       | L   | Ø A <sup>-0.005</sup> <sub>-0.020</sub> | Ø B | C | Point Ø          |
|------------|-----|-----------------------------------------|-----|---|------------------|
| TOX 04.130 | 30  | 4                                       | 6   | 3 | 2                |
| TOX 10.25  | 60  | 10                                      | 13  | 5 | 3 / 4 / 5 / (6)* |
| TOX 10.180 | 80  | 10                                      | 13  | 5 | 3 / 4 / 5 / (6)* |
| TOX 10.200 | 100 | 10                                      | 13  | 5 | 3 / 4 / 5 / (6)* |
| TOX 12.25  | 60  | 12                                      | 15  | 5 | 6                |
| TOX 14.25  | 60  | 14                                      | 17  | 5 | 8                |
| TOX 14.180 | 80  | 14                                      | 17  | 5 | 8                |
| TOX 14.200 | 100 | 14                                      | 17  | 5 | 8                |
| TOX 16.25  | 60  | 16                                      | 19  | 5 | 10               |
| TOX 20.25  | 60  | 20                                      | 23  | 5 | 12               |

\* only after consulting TOX® PRESSOTECHNIK

Dimensions in mm

### Special shapes of flanged tools



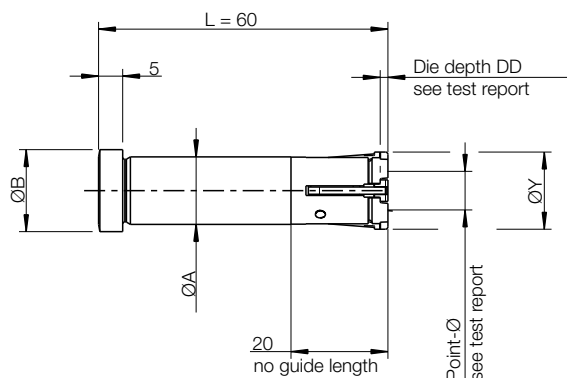
### Tapers

For piece parts with small flange widths or for components which require a penetration of the tool, flanged tools with tapered flange can be used.

Please note that, due to the press forces, the design is limited. Each tool with special shape has to be analyzed in the test laboratory.

## Flanged Tools

### TOX® SKB flanged die



| Type       | L   | Ø A <sup>-0.005<br/>-0.020</sup> | Ø B | Point Ø | Y**  |
|------------|-----|----------------------------------|-----|---------|------|
| SKB 10.25  | 60  | 10                               | 13  | 5 - 6   | 12   |
| SKB 12.25  | 60  | 12                               | 15  | 6 - 7   | 14   |
| SKB 14.25  | 60  | 14                               | 17  | 8 - 9   | 16   |
| SKB 14.180 | 80  | 14                               | 17  | 8 - 9   | 16   |
| SKB 14.200 | 100 | 14                               | 17  | 8 - 9   | 16   |
| SKB 14.220 | 120 | 14                               | 17  | 8 - 9   | 16   |
| SKB 16.25* | 60  | 16                               | 19  | 8 - 10  | 17.5 |

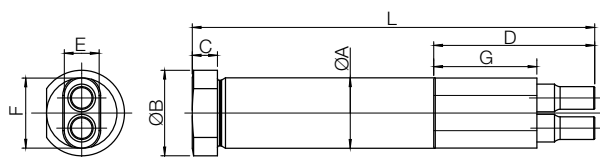
\* Joining with intermediate layer (e.g. glue/textiles)

Dimensions in mm

\*\* = with max. opened die

### TOX® TWINpoint flanged punch

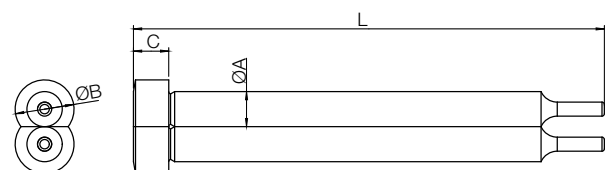
Type TOX 210, 214, 218



| Type        | L   | Ø A <sup>-0.005<br/>-0.020</sup> | Ø B | C | D  | E  | F  | G    | Point Ø |
|-------------|-----|----------------------------------|-----|---|----|----|----|------|---------|
| TOX 210.180 | 80  | 10                               | 13  | 5 | 28 | 6  | 10 | 14.5 | 4       |
| TOX 210.200 | 100 | 10                               | 13  | 5 | 28 | 6  | 10 | 14.5 | 4       |
| TOX 210.220 | 120 | 10                               | 13  | 5 | 28 | 6  | 10 | 14.5 | 4       |
| TOX 214.180 | 80  | 14                               | 17  | 5 | 28 | 7  | 12 | 14.5 | 5       |
| TOX 214.180 | 80  | 14                               | 17  | 5 | 32 | 8  | 14 | 20.5 | 6       |
| TOX 214.200 | 100 | 14                               | 17  | 5 | 28 | 7  | 12 | 14.5 | 5       |
| TOX 214.200 | 100 | 14                               | 17  | 5 | 32 | 8  | 14 | 20.5 | 6       |
| TOX 214.220 | 120 | 14                               | 17  | 5 | 28 | 7  | 12 | 14.5 | 5       |
| TOX 214.220 | 120 | 14                               | 17  | 5 | 32 | 8  | 14 | 20.5 | 6       |
| TOX 218.200 | 100 | 18                               | 21  | 5 | 32 | 10 | 18 | 20.5 | 8       |
| TOX 218.220 | 120 | 18                               | 21  | 5 | 32 | 10 | 18 | 20.5 | 8       |

Dimensions in mm

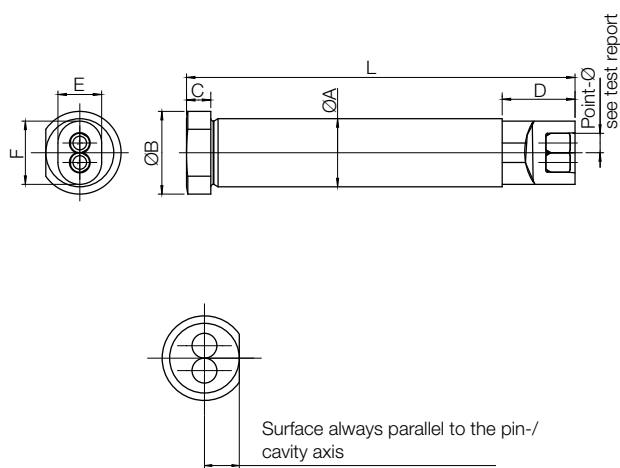
Type 2 x TOX



| Type           | L   | Ø A <sup>-0.005<br/>-0.020</sup> | Ø B | C | D | E | F | G | Point Ø |
|----------------|-----|----------------------------------|-----|---|---|---|---|---|---------|
| 2 x TOX 03.141 | 40  | 3                                | 5   | 3 | - | - | - | - | 3       |
| 2 x TOX 10.205 | 100 | 10                               | 13  | 5 | - | - | - | - | 10      |
| 2 x TOX 10.227 | 120 | 10                               | 13  | 5 | - | - | - | - | 10      |

Dimensions in mm

### TOX® TWINpoint flanged die



| Type        | L   | Ø A <sup>-0.005<br/>-0.020</sup> | Ø B | C | D  | E  | F  | Point Ø |
|-------------|-----|----------------------------------|-----|---|----|----|----|---------|
| TOX 210.160 | 60  | 10                               | 13  | 5 | 15 | 7  | 10 | 3       |
| TOX 214.180 | 80  | 14                               | 17  | 5 | 15 | 9  | 13 | 4       |
| TOX 216.180 | 80  | 16                               | 19  | 5 | 20 | 10 | 15 | 5       |
| TOX 218.180 | 80  | 18                               | 21  | 5 | 20 | 12 | 18 | 6       |
| TOX 222.200 | 100 | 22                               | 25  | 5 | 20 | 14 | 22 | 8       |
| TOX 226.220 | 120 | 26                               | 29  | 5 | 20 | 16 | 26 | 10      |

Dimensions in mm

### Anti-rotation device

TOX® TWINpoint tools are secured against rotating by default. The surface on the flange which is in parallel to the pin-/cavity axis ensures the tool can not rotate inside the tool fixture. Suitable strippers can be found on page 8 and 9.

# Dies

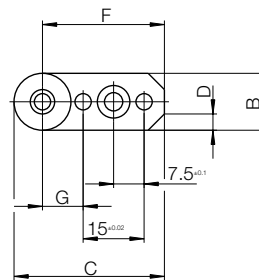
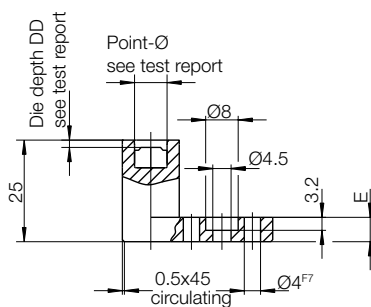
## Dies, offset

The offset die is particularly useful where access to the piece part is difficult, e.g. C- and U-shaped sections, blanks with 90° offset, etc.

The tools can be used for single or multipoint applications. Special shapes are available on request.

The tools can be replaced quickly and easily.

### TOX® Round Joint Die, offset

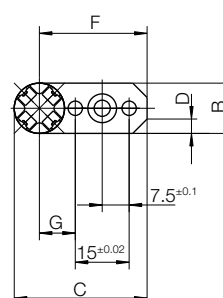
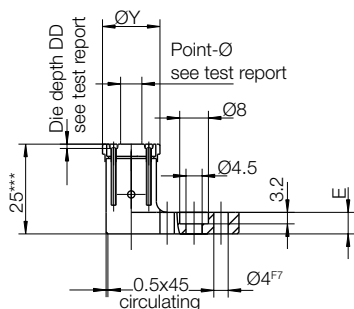


| Type      | B  | C  | D       | E  | F±0.02 | G±0.02 | Point Ø          |
|-----------|----|----|---------|----|--------|--------|------------------|
| TOX 50.25 | 10 | 35 | 3x45°   | 6  | 30     | 10     | 3 / 4 / 5 / (6)* |
| TOX 51.22 | 12 | 36 | 3x45°   | 6  | 30     | 10     | 6                |
| TOX 51.25 | 14 | 37 | 4x45°   | 6  | 30     | 10     | 8                |
| TOX 53.25 | 16 | 38 | 4.5x45° | 10 | 30     | 10     | 10               |
| TOX 54.25 | 20 | 45 | 4.5x45° | 10 | 35     | 15     | 12               |

\* only after consulting TOX® PRESSTECHNIK

Dimensions in mm

### TOX® SKB Die, offset



| Type       | B  | C  | D       | E  | F±0.02 | G±0.02 | Point Ø | Y**  |
|------------|----|----|---------|----|--------|--------|---------|------|
| SKB 50.25  | 10 | 35 | 3x45°   | 6  | 30     | 10     | 5 – 6   | 12   |
| SKB 51.22  | 12 | 36 | 3x45°   | 6  | 30     | 10     | 6 – 7   | 14   |
| SKB 51.25  | 14 | 37 | 4x45°   | 6  | 30     | 10     | 8 – 9   | 16   |
| SKB 53.25* | 16 | 38 | 4.5x45° | 10 | 30     | 10     | 8 – 10  | 17.5 |

\* joining with intermediate layer (e.g. glue/textiles)

\*\* = with max. opened die

\*\*\* minimum dimension 16 mm, deliverable upon request

Dimensions in mm

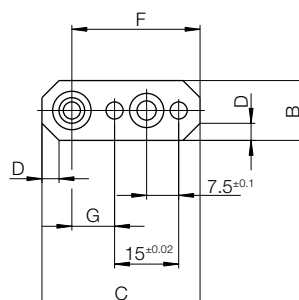
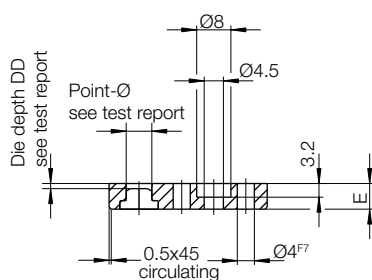
## Dies, flat

The flat die plate reaches into almost every corner of the piece part. The shape allows for smaller flanges and extremely compact design.

The tools can be used for single or multipoint applications. Special shapes are available on request.

The tools can be replaced quickly and easily.

### TOX® Round Joint Die, flat



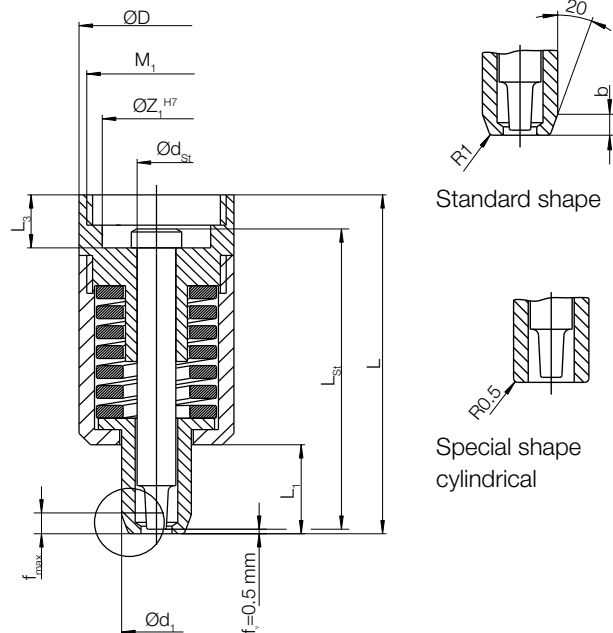
| Type      | B  | C  | D       | E  | F±0.02 | G±0.02 | Point Ø          |
|-----------|----|----|---------|----|--------|--------|------------------|
| TOX 40.25 | 10 | 35 | 3x45°   | 6  | 30     | 10     | 3 / 4 / 5 / (6)* |
| TOX 41.01 | 12 | 36 | 3x45°   | 6  | 30     | 10     | 6                |
| TOX 41.25 | 14 | 37 | 4x45°   | 6  | 30     | 10     | 8                |
| TOX 43.25 | 16 | 38 | 4.5x45° | 10 | 30     | 10     | 10               |
| TOX 44.25 | 20 | 45 | 4.5x45° | 10 | 35     | 15     | 12               |

\* only after consulting TOX® PRESSOTECHNIK

Dimensions in mm

# Stripper

## Punch Stripper, round



### Punch stripper CSR for TOX® Flanged Punches

For single and multipoint solutions, for holding flanges type CZP, CZE and CZW and for drives RZK, RZH, EPMR.

The punch stripper consist of punch holder, stripping sleeve, spring sleeve and special helical spring.

$L_{St}$  = Length of punch

$d_{St}$  = Diameter of punch

$F_v$  = Preload force

$f_v$  = Punch recessed

A description of how to calculate the punch stripper correctly can be found in data sheet 80.18 „Installation and design guidelines“.

### Punch stripper suitable for holding flanges CZP, CZW, CZE

| Type          | $L_{St}$ | $d_{St}$ | $F_v(N)$ | Rate of spring<br>$R(N/mm)$ | $\varnothing D$ | $L$   | $\varnothing d_1$ | $b$ | $\varnothing Z_1^{H7}$ | $M_1$   | $L_1$ | $L_3$ | Spring type | Recommended die stripper |
|---------------|----------|----------|----------|-----------------------------|-----------------|-------|-------------------|-----|------------------------|---------|-------|-------|-------------|--------------------------|
| CSR 16.03.040 | 40       | 3        | 300      | 250                         | 30              | 48    | 7                 | —   | 18                     | M20x1.5 | 13    | 10    | SZ8590.016  |                          |
| CSR 25.10.060 | 60       | 10       | 925      | 370                         | 32              | 67.5  | 15                | 2.5 | 22                     | M24x1.5 | 11    | 12    | CZF25       | CMR 24...                |
| CSR 25.10.080 | 80       | 10       | 925      | 370                         | 32              | 87.5  | 15                | 2.5 | 22                     | M24x1.5 | 31    | 12    | CZF25       | CMR 24...                |
| CSR 25.10.100 | 100      | 10       | 925      | 370                         | 32              | 107.5 | 15                | 2.5 | 22                     | M24x1.5 | 51    | 12    | CZF25       | CMR 24...                |
| CSR 32.10.080 | 80       | 10       | 970      | 485                         | 40              | 88.5  | 18                | 5   | 28                     | M32x1.5 | 23.5  | 13    | CZF32       | CMR 24...                |
| CSR 32.10.100 | 100      | 10       | 970      | 485                         | 40              | 108.5 | 18                | 5   | 28                     | M32x1.5 | 43.5  | 13    | CZF32       | CMR 31...                |
| CSR 38.10.080 | 80       | 10       | 2610     | 580                         | 50              | 90.5  | 22                | 7   | 28                     | M42x1.5 | 12.5  | 15    | CZF38       | CMR 24.../31...          |
| CSR 38.10.100 | 100      | 10       | 2610     | 580                         | 50              | 110.5 | 22                | 7   | 28                     | M42x1.5 | 32.5  | 15    | CZF38       | CMR 32.../37...          |
| CSR 51.10.100 | 100      | 10       | 4020     | 670                         | 65              | 110.5 | 24                | 7   | 40                     | M56x1.5 | 19.5  | 15    | CZF51       | alle CMR                 |

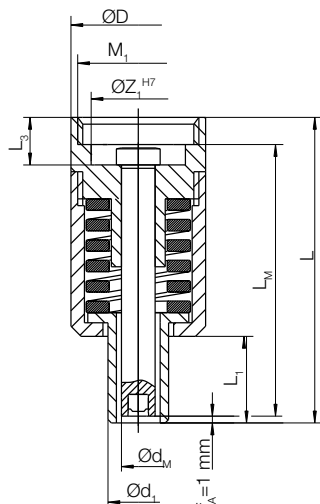
Dimensions in mm

### Punch stripper suitable for drives RZK, RZH, EPMR

| Type           | $L_{St}$ | $d_{St}$ | $F_v(N)$ | Rate of spring<br>$R(N/mm)$ | $\varnothing D$ | $L$   | $\varnothing d_1$ | $b$ | $\varnothing Z_1^{H7}$ | $M_1$   | $L_1$ | $L_3$ | Spring type | Recommended die stripper |
|----------------|----------|----------|----------|-----------------------------|-----------------|-------|-------------------|-----|------------------------|---------|-------|-------|-------------|--------------------------|
| CSR 025.10.060 | 60       | 10       | 925      | 370                         | 32              | 75.5  | 18                | 2.5 | 22                     | M24x1.5 | 11    | 20.05 | CZF25       | CMR 024                  |
| CSR 025.10.080 | 80       | 10       | 925      | 370                         | 32              | 95.5  | 18                | 2.5 | 22                     | M24x1.5 | 31    | 20.05 | CZF25       | CMR 024                  |
| CSR 025.10.100 | 100      | 10       | 925      | 370                         | 32              | 110.5 | 18                | 2.5 | 22                     | M24x1.5 | 51    | 20.05 | CZF25       | CMR 024...               |
| CSR 032.10.080 | 80       | 10       | 970      | 485                         | 40              | 95.5  | 18                | 5   | 22                     | M24x1.5 | 23.5  | 20.05 | CZF32       | CMR 024/030              |
| CSR 032.10.100 | 100      | 10       | 970      | 485                         | 40              | 115.5 | 18                | 5   | 22                     | M24x1.5 | 43.5  | 20.05 | CZF32       | CMR 024/030              |
| CSR 032.10.120 | 120      | 10       | 970      | 485                         | 40              | 135.5 | 18                | 5   | 22                     | M24x1.5 | 63.5  | 20.05 | CZF32       | CMR 024/030              |
| CSR 038.10.080 | 80       | 10       | 2610     | 580                         | 50              | 95.5  | 22                | 7   | 22                     | M24x1.5 | 12.5  | 20.05 | CZF38       | CMR 030/031              |
| CSR 038.10.100 | 100      | 10       | 2610     | 580                         | 50              | 115.5 | 22                | 7   | 22                     | M24x1.5 | 32.5  | 20.05 | CZF38       | CMR 030/031              |
| CSR 038.10.120 | 120      | 10       | 2610     | 580                         | 50              | 135.5 | 22                | 7   | 22                     | M24x1.5 | 52.5  | 20.05 | CZF38       | CMR 030/031              |
| CSR 051.10.100 | 100      | 10       | 4020     | 670                         | 65              | 115.5 | 24                | 7   | 22                     | M24x1.5 | 19.5  | 20.05 | CZF51       | alle CMR                 |
| CSR 051.10.120 | 120      | 10       | 4020     | 670                         | 65              | 135.5 | 24                | 7   | 22                     | M24x1.5 | 39.5  | 20.05 | CZF51       | alle CMR                 |

Dimensions in mm

## Die Stripper, round



### Die stripper CMR for TOX® Flanged Dies

Die holder, for single and multipoint solutions, installed directly on plate with holding flanges type CZE and CZW and for mounting to TOX® Powerpackage with holding flange type CZP. The die stripper consists of die holder, stripping sleeve, spring sleeve and special helical spring.

$L_m$  = Length of die

$d_m$  = Diameter of die

$F_v$  = Preload force

$F$  = Stripping force

$f_A$  = Stripping travel

A description of how to calculate the die stripper correctly can be found in data sheet 80.18 „Installation and design guidelines“.

### Die stripper suitable for holding flanges CZP, CZW, CZE

| Type          | $L_m$ | $d_m$ | $F_v$ (N) | Rate of spring R<br>(N/mm) | $\varnothing D$ | L   | $\varnothing d_1$ | $\varnothing Z_1^{H7}$ | $M_1$   | $L_1$ | $L_3$ | Spring type |
|---------------|-------|-------|-----------|----------------------------|-----------------|-----|-------------------|------------------------|---------|-------|-------|-------------|
| CMR 16.04.030 | 30    | 4     | 300       | 250                        | 30              | 38  | 10                | 18                     | M20x1.5 | 3     | 10    | SZ8590.016  |
| CMR 24.14.060 | 60    | 14    | 900       | 95                         | 32              | 68  | 19                | 22                     | M24x1.5 | 11.5  | 12    | CZF24       |
| CMR 24.14.080 | 80    | 14    | 900       | 95                         | 32              | 88  | 19                | 22                     | M24x1.5 | 31.5  | 12    | CZF24       |
| CMR 24.14.100 | 100   | 14    | 900       | 95                         | 32              | 108 | 19                | 22                     | M24x1.5 | 51.5  | 12    | CZF24       |
| CMR 25.10.060 | 60    | 10    | 2035      | 370                        | 32              | 68  | 15                | 22                     | M24x1.5 | 11.5  | 12    | CZF25       |
| CMR 25.10.080 | 80    | 10    | 2035      | 370                        | 32              | 88  | 15                | 22                     | M24x1.5 | 31.5  | 12    | CZF25       |
| CMR 25.10.100 | 100   | 10    | 2035      | 370                        | 32              | 108 | 15                | 22                     | M24x1.5 | 51.5  | 12    | CZF25       |
| CMR 31.14.080 | 80    | 14    | 2275      | 350                        | 40              | 89  | 18                | 28                     | M32x1.5 | 20    | 13    | CZF31       |
| CMR 31.14.100 | 100   | 14    | 2275      | 350                        | 40              | 109 | 18                | 28                     | M32x1.5 | 40    | 13    | CZF31       |
| CMR 32.14.080 | 80    | 14    | 3150      | 485                        | 40              | 89  | 18                | 28                     | M32x1.5 | 20    | 13    | CZF32       |
| CMR 32.14.100 | 100   | 14    | 3150      | 485                        | 40              | 109 | 18                | 28                     | M32x1.5 | 40    | 13    | CZF32       |
| CMR 37.14.080 | 80    | 14    | 3410      | 325                        | 50              | 91  | 22                | 28                     | M42x1.5 | 9     | 15    | CZF37       |
| CMR 37.14.100 | 100   | 14    | 3410      | 325                        | 50              | 111 | 22                | 28                     | M42x1.5 | 29    | 15    | CZF37       |
| CMR 38.14.080 | 80    | 14    | 6090      | 580                        | 50              | 91  | 22                | 28                     | M42x1.5 | 9     | 15    | CZF38       |
| CMR 38.14.100 | 100   | 14    | 6090      | 580                        | 50              | 111 | 22                | 28                     | M42x1.5 | 29    | 15    | CZF38       |

Dimensions in mm

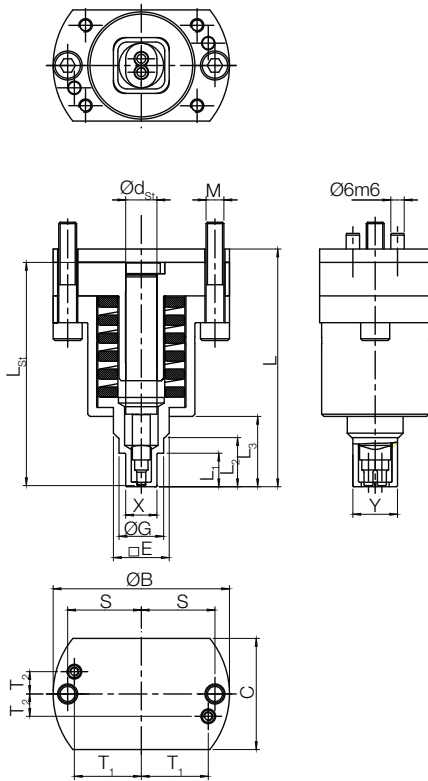
### Die stripper suitable for drives RZK, RZH, EPMR

| Type           | $L_m$ | $d_m$ | $F_v$ (N) | Rate of spring R<br>(N/mm) | $\varnothing D$ | L   | $\varnothing d_1$ | $\varnothing Z_1^{H7}$ | $M_1$   | $L_1$ | $L_3$ | Spring type |
|----------------|-------|-------|-----------|----------------------------|-----------------|-----|-------------------|------------------------|---------|-------|-------|-------------|
| CMR 024.14.060 | 60    | 14    | 900       | 95                         | 32              | 76  | 19                | 22                     | M24x1.5 | 11.5  | 20.05 | CZF24       |
| CMR 024.14.080 | 80    | 14    | 900       | 95                         | 32              | 96  | 19                | 22                     | M24x1.5 | 31.5  | 20.05 | CZF24       |
| CMR 025.10.060 | 60    | 10    | 2035      | 370                        | 32              | 76  | 15                | 22                     | M24x1.5 | 11.5  | 20.05 | CZF25       |
| CMR 025.10.080 | 80    | 10    | 2035      | 370                        | 32              | 96  | 15                | 22                     | M24x1.5 | 31.5  | 20.05 | CZF25       |
| CMR 031.14.080 | 80    | 14    | 2258      | 350                        | 40              | 96  | 18                | 28                     | M24x1.5 | 24    | 20.05 | CZF31       |
| CMR 031.14.100 | 100   | 14    | 2258      | 350                        | 40              | 116 | 18                | 28                     | M24x1.5 | 44    | 20.05 | CZF31       |
| CMR 032.14.080 | 80    | 14    | 3150      | 485                        | 40              | 96  | 18                | 28                     | M24x1.5 | 24    | 20.05 | CZF32       |
| CMR 032.14.100 | 100   | 14    | 3150      | 485                        | 40              | 116 | 18                | 28                     | M24x1.5 | 44    | 20.05 | CZF32       |
| CMR 037.14.080 | 80    | 14    | 3735      | 325                        | 50              | 96  | 22                | 28                     | M24x1.5 | 13    | 20.05 | CZF37       |
| CMR 037.14.100 | 100   | 14    | 3735      | 325                        | 50              | 116 | 22                | 28                     | M24x1.5 | 33    | 20.05 | CZF37       |
| CMR 038.14.080 | 80    | 14    | 6670      | 580                        | 50              | 96  | 22                | 28                     | M24x1.5 | 13    | 20.05 | CZF38       |
| CMR 038.14.100 | 100   | 14    | 6670      | 580                        | 50              | 116 | 22                | 28                     | M24x1.5 | 33    | 20.05 | CZF38       |

Dimensions in mm

# Stripper

## Punch Stripper TWINpoint



### Punch stripper CSRT for TOX® TWINpoint Punch

For single and multipoint solutions, installed directly on plate.

The punch stripper consists of punch holder, stripping sleeve, spring sleeve and special helical spring.

$L_{st}$  = Length of punch

$d_{st}$  = Diameter of punch

$F_v$  = Preload force

$f_v$  = Punch recessed

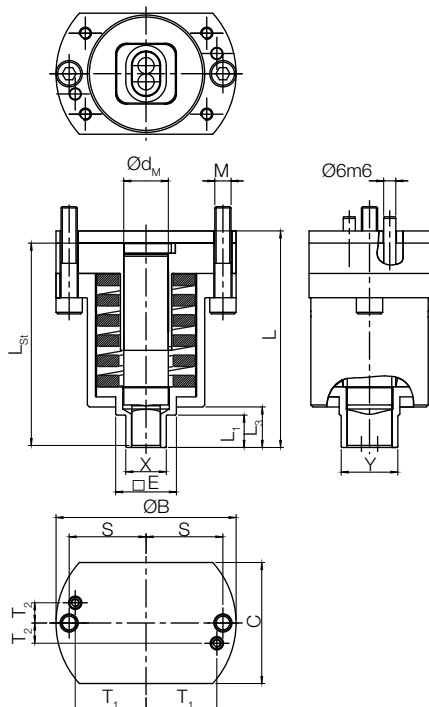
A description of how to calculate the punch stripper correctly can be found in data sheet 80.18 „Installation and design guidelines“.

| Type                     | L <sub>St</sub> | d <sub>St</sub> | F <sub>V</sub> (N) | Rate of spring |  | Ø B | C  | L    | M  | X  | Y  | E  | Ø G | L <sub>1</sub> | L <sub>2</sub> | L <sub>3</sub> | T <sub>1</sub> | T <sub>2</sub> | S  | Point Ø | Spring type |
|--------------------------|-----------------|-----------------|--------------------|----------------|--|-----|----|------|----|----|----|----|-----|----------------|----------------|----------------|----------------|----------------|----|---------|-------------|
|                          |                 |                 |                    | R (N/mm)       |  |     |    |      |    |    |    |    |     |                |                |                |                |                |    |         |             |
| CSRT 32.10.04.080.035.00 | 80              | 10              | 970                | 485            |  | 71  | 42 | 86.5 | M8 | 10 | 14 | 21 |     | 18             |                | 20.5           | 29             | 9              | 29 | 4       | CZF32       |
| CSRT 32.10.04.100.035.00 | 100             | 10              | 970                | 485            |  | 71  | 42 | 107  | M8 | 10 | 14 | 21 | 15  | 18             | 30             | 40.5           | 29             | 9              | 29 | 4       | CZF32       |
| CSRT 32.10.04.120.035.00 | 120             | 10              | 970                | 485            |  | 71  | 42 | 127  | M8 | 10 | 14 | 21 | 15  | 18             | 50             | 60.5           | 29             | 9              | 29 | 4       | CZF32       |
| CSRT 38.14.06.080.065.00 | 80              | 14              | 2610               | 580            |  | 79  | 50 | 86.5 | M8 | 14 | 20 | 25 | –   | 8              | –              | 10.5           | 30             | 10             | 33 | 6 (5)   | CZF38       |
| CSRT 38.14.06.100.065.00 | 100             | 14              | 2610               | 580            |  | 79  | 50 | 107  | M8 | 14 | 20 | 25 | –   | 14             | –              | 30.5           | 30             | 10             | 33 | 6 (5)   | CZF38       |
| CSRT 38.14.06.120.065.00 | 120             | 14              | 2610               | 580            |  | 79  | 50 | 127  | M8 | 14 | 20 | 25 | –   | 14             | –              | 50.5           | 30             | 10             | 33 | 6 (5)   | CZF38       |
| CSRT 51.18.08.100.075.00 | 100             | 18              | 4020               | 670            |  | 89  | 60 | 107  | M8 | 15 | 23 | 30 | –   | 9              | –              | 19.5           | 35             | 10             | 38 | 8       | CZF51       |
| CSRT 51.18.08.120.075.00 | 120             | 18              | 4020               | 670            |  | 89  | 60 | 127  | M8 | 15 | 23 | 30 | –   | 19             | –              | 39.5           | 35             | 10             | 38 | 8       | CZF51       |
| CSRT 51.10.10.100.100.00 | 100             | 2x10            | 4020               | 670            |  | 89  | 60 | 107  | M8 | 18 | 18 | 30 | –   | 11             | –              | 19.5           | 35             | 10             | 38 | 10      | CZF51       |
| CSRT 51.10.10.120.100.00 | 120             | 2x10            | 4020               | 670            |  | 89  | 60 | 127  | M8 | 18 | 28 | 30 | –   | 31             | –              | 39.5           | 35             | 10             | 38 | 10      | CZF51       |

Dimensions in mm



## Die Stripper TWINpoint



### Die stripper CMRT for TOX® TWINpoint Flanged Dies

Die stripper, for single and multi-point solutions, with die holder installed directly on plate.

The die stripper consists of die holder, stripping sleeve, spring sleeve and special helical spring.

$L_M$  = Length of die

$d_M$  = Diameter of die

$F_v$  = Preload force

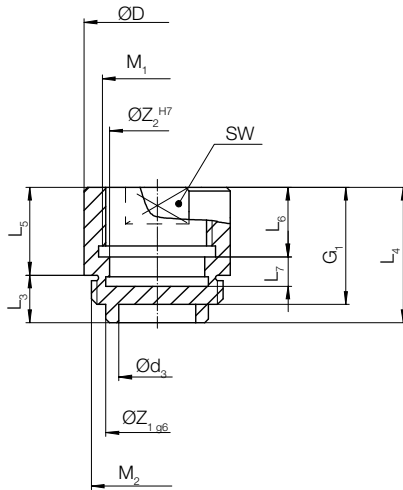
$f_v$  = Die recessed

A description of how to calculate the die stripper correctly can be found in data sheet 80.18 „Installation and design guidelines“.

| Type                 | $L_M$ | $d_M$ | Rate of spring |          | $\varnothing B$ | C  | L   | M  | X  | Y  | E  | $L_1$ | $L_2$ | $T_1$ | $T_2$ | S    | Point $\varnothing$ | Spring type |
|----------------------|-------|-------|----------------|----------|-----------------|----|-----|----|----|----|----|-------|-------|-------|-------|------|---------------------|-------------|
|                      |       |       | $F_v$ (N)      | R (N/mm) |                 |    |     |    |    |    |    |       |       |       |       |      |                     |             |
| CMRT 30.14.04.080.00 | 80    | 14    | 680            | 170      | 71              | 42 | 87  | M8 | 14 | 18 | 21 | 12    | 21    | 29    | 9     | 29   | 4                   | CZF30       |
| CMRT 37.18.06.080.00 | 80    | 18    | 1788           | 325      | 79              | 50 | 87  | M8 | 18 | 24 | 25 | 10    | 11    | 30    | 10    | 33   | 6 (5)               | CZF37       |
| CMRT 50.22.08.100.00 | 100   | 22    | 2310           | 385      | 89              | 60 | 107 | M8 | 20 | 28 | 30 | 16    | 20    | 30    | 10    | 38   | 8                   | CZF50       |
| CMRT 62.26.10.120.00 | 120   | 26    | 3150           | 630      | 102             | 73 | 127 | M8 | 22 | 32 | 34 | 15    | 27    | 35    | 15    | 44.5 | 10                  | CZF62       |

Dimensions in mm

# Holding Flanges

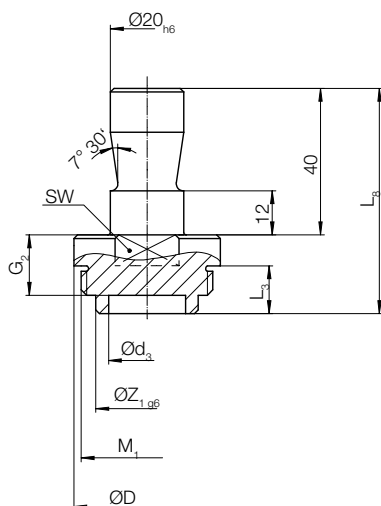


## Holding flange CZP

Holding flange for stripper type CSR and CMR, for screwing onto the TOX® Powerpackage working piston.

| Type                | ØD <sub>-0.2</sub> | L <sub>3</sub> | L <sub>4</sub> | L <sub>5</sub> | L <sub>6</sub> | L <sub>7</sub> | M <sub>1</sub> | M <sub>2</sub> | Z <sub>1 g6</sub> | ØZ <sub>2 H7</sub> | Ød <sub>3</sub> | G <sub>1</sub> | SW |
|---------------------|--------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-------------------|--------------------|-----------------|----------------|----|
| CZP 022.024.026.000 | 31.8               | 12             | 38             | 26             | 14             | 7              | M22x2          | M24x1.5        | 22                | 18                 | 18              | 33             | 27 |
| CZP 030.024.032.001 | 39.8               | 12             | 44             | 32             | 19             | 8              | M30x2          | M24x1.5        | 22                | 26                 | 18              | 39             | 36 |
| CZP 022.032.023.000 | 39.8               | 13             | 36             | 23             | 14             | 7              | M22x2          | M32x1.5        | 28                | 18                 | 21              | 31             | 36 |
| CZP 030.032.032.000 | 39.8               | 13             | 45             | 32             | 19             | 8              | M30x2          | M32x1.5        | 28                | 26                 | 21              | 40             | 36 |
| CZP 030.042.027.000 | 49.8               | 15             | 42             | 27             | 19             | 8              | M30x2          | M42x1.5        | 28                | 26                 | 21              | 37             | 46 |
| CZP 030.056.029.000 | 64.8               | 15             | 44             | 29             | 19             | 8              | M30x2          | M56x1.5        | 40                | 26                 | 25              | 39             | 60 |

Dimensions in mm

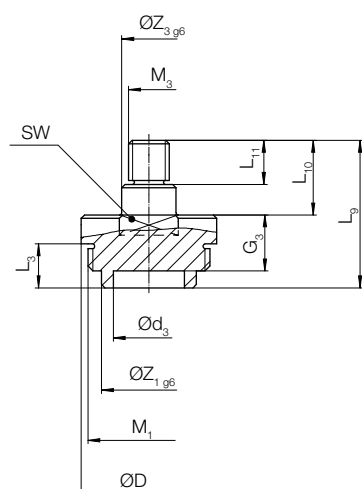


## Holding flange CZW

Holding flange for stripper type CSR and CMR, adaptable on plate side.

| Type                | L <sub>3</sub> | L <sub>8</sub> | ØD <sub>-0.2</sub> | M <sub>1</sub> | Z <sub>1 g6</sub> | Ød <sub>3</sub> | G <sub>2</sub> | SW |
|---------------------|----------------|----------------|--------------------|----------------|-------------------|-----------------|----------------|----|
| CZW 020.024.008.000 | 12             | 60.5           | 31.8               | M24x1.5        | 22                | 18              | 15.5           | 27 |
| CZW 020.032.008.000 | 13             | 61.5           | 39.8               | M32x1.5        | 28                | 21              | 16.5           | 36 |
| CZW 020.042.008.000 | 15             | 63.5           | 49.8               | M42x1.5        | 28                | 21              | 18.5           | 46 |
| CZW 020.056.009.000 | 15             | 64.5           | 64.8               | M56x1.5        | 40                | 25              | 19.5           | 60 |
| CZW 020.024.009.000 | 19             | 67.5           | 32                 | M24x1.5        | 22                | 17.3            | 23.5           | 30 |

Dimensions in mm



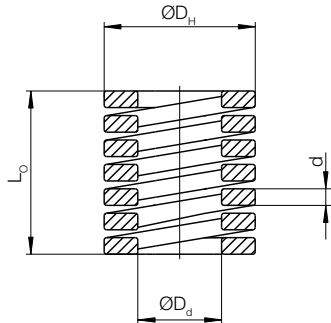
### Holding flange CZE

Holding flange for stripper type CSR and CMR, for screwing into plate.

| Type                | ØD <sub>-0.2</sub> | L <sub>3</sub> | L <sub>9</sub> | L <sub>10</sub> | L <sub>11</sub> | M <sub>1</sub> | M <sub>3</sub> | Z <sub>1g6</sub> | ØZ <sub>3g6</sub> | Ød <sub>3</sub> | G <sub>3</sub> | SW |
|---------------------|--------------------|----------------|----------------|-----------------|-----------------|----------------|----------------|------------------|-------------------|-----------------|----------------|----|
| CZE 012.024.009.000 | 31.8               | 12             | 43.0           | 22              | 13              | M24x1.5        | M12x1.5        | 22               | 16                | 18              | 16.0           | 27 |
| CZE 012.032.008.000 | 39.8               | 13             | 43.5           | 22              | 13              | M32x1.5        | M12x1.5        | 28               | 16                | 21              | 16.5           | 36 |
| CZE 012.042.008.000 | 49.8               | 15             | 45.5           | 22              | 13              | M42x1.5        | M12x1.5        | 28               | 16                | 21              | 18.5           | 46 |
| CZE 020.056.009.000 | 64.8               | 15             | 49.5           | 22              | 15              | M56x1.5        | M20x1.5        | 40               | 24                | 25              | 19.5           | 60 |
| CZE 012.024.012.001 | 39.8               | 19             | 53.0           | 22              | 13              | M24x1.5        | M12x1.5        | 22               | 16                | 17.5            | 27.0           | 36 |

Dimensions in mm

# Special Helical Springs



## Special helical spring CZF

Special helical spring suitable for stripper type CSR and type CMR, ground to parallel and angular requirements.

$\varnothing D_H$  = Sleeve diameter (+ 0.2 mm tolerance)

$\varnothing D_d$  = Mandrel diameter (- 0.2 mm tolerance)

$L_0$  = Free length

$d$  = Wire diameter

$S_n$  = Spring travel

$F$  = Force

$R$  = Rate of spring ( $\pm 10\%$  tolerance)

| Type   | Ø D <sub>H</sub> | Ø D <sub>d</sub> | L <sub>0</sub> | d   | Rate of spring R<br>(N/mm) | Long lifetime | Average lifetime |                     | Max. travel / force |      | Color |        |
|--------|------------------|------------------|----------------|-----|----------------------------|---------------|------------------|---------------------|---------------------|------|-------|--------|
|        |                  |                  |                |     |                            | 60 % Sn       | 80 % Sn          | Sn F <sub>max</sub> |                     |      |       |        |
|        |                  |                  |                |     | (N/mm)                     | (mm)          | F (N)            | (mm)                | F (N)               | (mm) | (N)   |        |
| CZF 24 | 25               | 15.2             | 33             | 4.1 | 93                         | 6.3           | 586              | 8.4                 | 781                 | 10.5 | 977   | –      |
| CZF 22 | 25               | 12.5             | 31             | 3.1 | 110                        | 6.7           | 737              | 9.0                 | 990                 | 11.2 | 1232  | blue   |
| CZF 23 | 25               | 12.5             | 31             | 4.1 | 275                        | 5.8           | 1595             | 7.7                 | 2118                | 9.6  | 2640  | red    |
| CZF 25 | 25               | 12.5             | 31             | 4.6 | 370                        | 4.8           | 1776             | 6.4                 | 2368                | 8.0  | 2960  | yellow |
| CZF 30 | 32               | 16               | 37             | 4.0 | 170                        | 8.0           | 1360             | 10.6                | 1802                | 13.3 | 2261  | blue   |
| CZF 31 | 32               | 16               | 37             | 5.3 | 350                        | 6.8           | 2380             | 9.1                 | 3185                | 11.4 | 3990  | red    |
| CZF 32 | 32               | 16               | 37             | 5.6 | 485                        | 5.7           | 2765             | 7.6                 | 3686                | 9.5  | 4608  | yellow |
| CZF 36 | 40               | 20               | 49.5           | 4.8 | 170                        | 10.7          | 1819             | 14.3                | 2434                | 17.9 | 3043  | blue   |
| CZF 37 | 40               | 20               | 49.5           | 6.2 | 325                        | 9.2           | 2990             | 12.2                | 3965                | 15.3 | 4973  | red    |
| CZF 38 | 40               | 20               | 49.5           | 7.3 | 580                        | 7.7           | 4466             | 10.2                | 5916                | 12.8 | 7424  | yellow |
| CZF 49 | 50               | 25               | 62             | 6.0 | 195                        | 13.4          | 2613             | 17.9                | 3491                | 22.4 | 4370  | blue   |
| CZF 50 | 50               | 25               | 62             | 7.4 | 385                        | 11.5          | 4428             | 15.4                | 5929                | 19.2 | 7392  | red    |
| CZF 51 | 50               | 25               | 62             | 9.1 | 670                        | 9.6           | 6432             | 12.8                | 8576                | 16.0 | 10720 | yellow |
| CZF 62 | 63               | 38               | 74             | 11  | 630                        | 13.7          | 8618             | 18.2                | 11491               | 22.8 | 14364 | red    |

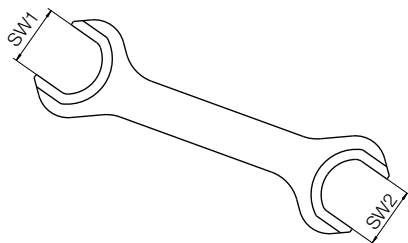
Dimensions in mm

Note:

To avoid any side forces on the TOX® Tool, no standard die springs must be used.

# Assembly tools

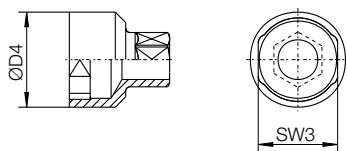
## Open end wrench CZG



Special tool for changing TOX® Tools, TOX® Helical Springs and holding flanges.

| Type         | SW 1 | SW 2 | appropriate for                                         |
|--------------|------|------|---------------------------------------------------------|
| CZG 27-30.00 | 27   | 30   | CZP/CZW/CZE 25<br>CSR/CMR 25/CMR 24                     |
| CZG 36-38.00 | 36   | 38   | CZP/CZW/CZE 32<br>CSR/CMR 32/CMR 31                     |
| CZG 46-60.00 | 46   | 60   | CSR/CMR/CZP/CZW/CZE 38/CMR 37<br>CSR/CMR/CZP/CZW/CZE 51 |

## Special hexagon socket CZN



| Type   | SW 3 | Ø D4 | appropriate for    |
|--------|------|------|--------------------|
| CZN 25 | 30   | 36.5 | CSR/CMR 25, CMR 24 |
| CZN 32 | 38   | 45.5 | CSR/CMR 32, CMR 31 |
| CZN 38 | 46   | 55.5 | CSR/CMR 38, CMR 37 |
| CZN 51 | 60   | 69.5 | CSR 51             |